Remote Sensing of Turbidity in Smaller Rivers

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Motivation

Turbidity is a key water quality parameter that is needed for environmental standards and domestic use. Suspend solids (SS) in water can have significant deleterious impacts on the physical, chemical and biological properties of the waterbody. Physical alterations caused by SS include reduced penetration of light, temperature changes, and infilling of channels and reservoirs when solids are deposited. Turbidity is a direct measure of the light-scattering properties of water, which is the primary factor affecting the ability of phytoplankton and macrophytes to photosynthesize and it accurately describes aesthetic issues and the reduction in light penetration, which are essential for recreational water use and basic environmental health assessments (Bilotta & Brazier, 2008). While conventional gravimetric methods provide high precision, they lack the spatiotemporal resolution required for effective watershed management. Remote sensing is a necessary alternative that offers a cost-effective and non-invasive method to monitor turbidity across vast and inaccessible river systems in real-time. (Dethier et al., 2020). This research focuses on the application of these methods to smaller transboundary rivers, such as the Olifants River, which is the main water source of the Limpopo region as well as Mozambique. The Olifants River is a vital water source, but as it runs through a natural park, the traditional method of "grab sampling" is not always possible, as it is disruptive to the local wildlife. With the advent of high-resolution satellite imagery, we are now able to survey this vital water source without disrupting the natural balance of the wildlife in the area.

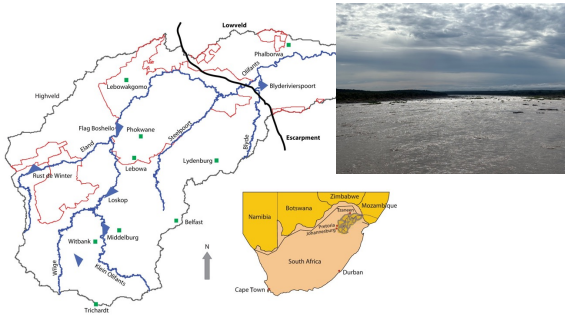

Figure 1: Olifants River (https://www.academia.edu/76192625/Evaluation_of_historic_current_and_future_water_demand_in_the_Olifants_River_Catchment_South_Africa)

Background

Current water quality monitoring is largely based on the concentration-response model, which assumes a predictable environmental impact relative to the level of SS. While global research has established turbidity as an excellent surrogate for these solids due to its direct measurement of light-scattering properties (Bilotta & Brazier, 2008) , many existing remote sensing models were originally developed and tested on large-scale water bodies or open oceans(Umar et al., 2018). These algorithms frequently fail when applied to smaller, optically complex river systems like the Olifants River (Fortsch and Kahler, 2024). Despite these failures, the core principle remains valid: suspended matter increases backscattering in a way that is detectable in the visible and near-infrared (NIR) spectrum. This study adapts these proven spectral principles but applies them to the unique, high-priority context of the Olifants River. By utilizing the high-resolution Sentinel-2 Multispectral imager (MSI).

Future Applications

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Methodology

Turbidity data were obtained from an in-situ turbidity sensor (PME Fixed-cyclops 7, Vista, CA, USA with Turner Designs Cyclops sensor, San Jose, CA, USA) and monitored by park scientific staff (Figure 2). Remote sensing data were obtained via Sentinel-2 Multispectral Imager (Google Earth Engine Data Catalog, European Space Agency). There are two Sentinal-2 satellites (A&B, Table 1) that cover the Earth every ten days with a combined revisit time of five days; the multispectral imager (MSI) collects 13 bands (Figure 3).

***Figure 2**: Installations of the the turbidity sensor on the Olifants River site*

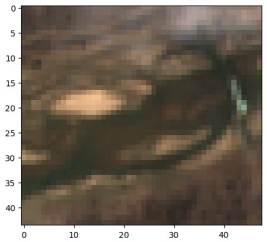

***Figure 3:** Composite red-green-blue (RGB) image of the Balule Weir from Sentinel-2 (August 2023).*

***Table 1:** Spectral bands for Sentinel-2 MSI (both A&B satellites) with resolution.*

Band Number	Sentinel-2A Central wavelength (nm)	Sentinel-2A Bandwidth (nm)	Sentinel-2B Central wavelength (nm)	Sentinel-2B Bandwidth (nm)	Spatial resolution (m)
1	442.7	20	442.3	20	60
2	492.7	65	492.3	65	10
3	559.8	35	558.9	35	10
4	664.6	30	664.9	31	10
5	704.1	14	703.8	15	20
6	740.5	14	739.1	13	20
7	782.8	19	779.7	19	20
8	832.8	105	832.9	104	10
8	864.7	21	864	21	20
9	945.1	19	943.2	20	60
10	1373.5	29	1376.9	29	60
11	1613.7	90	1610.4	94	20
12	2202.4	174	2185.7	184	20

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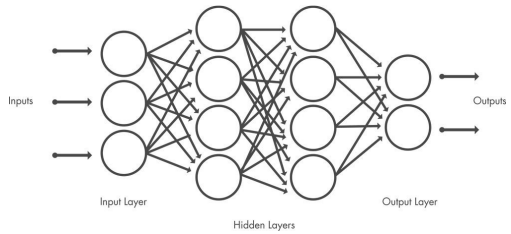
Convolutional Neural Network (CNN) was identified as the architecture network for deep learning ideal for finding patterns in images to recognize objects, classes, and categories.



Results

The model was trained with 20 imagines (Figure 4) and tested with the same 20 images. The plot of the predicted turbidity vs the measured turbidity (Figure 5) shows a linear relationship with an average error (RMSE) of 21.96 NTU

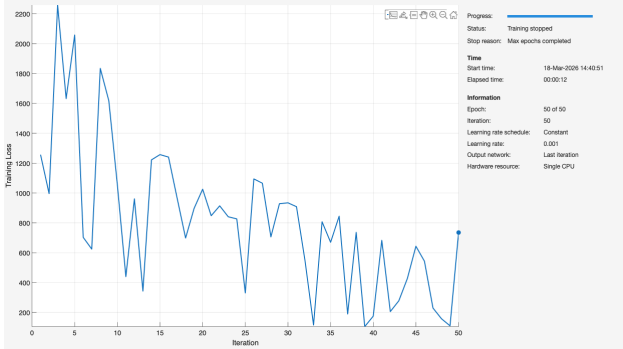


***Figure 4**: Training process*

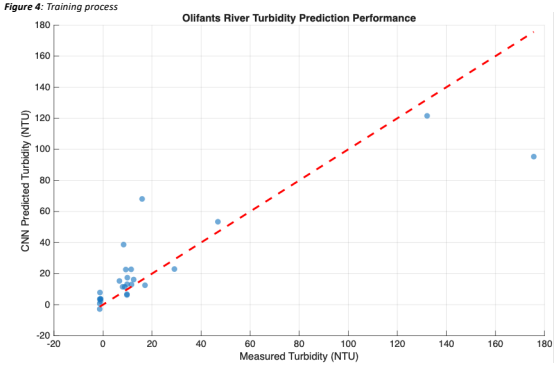


***Figure 5**: Plot of the predicted turbidity vs the measured turbidity*

Acknowledgements

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